

## CHAPTER 02

# Activities & Lab Packet

## Cells, Tissues & Body Systems

A complete set of labs, worksheets, writing prompts, discussions, and a group project to accompany Chapter 2. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S7L2

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## 1 Build-a-Cell Model

LAB

### OBJECTIVE

Build a 3-D or 2-D model of a cell and label every organelle with its job.

### TIME

40 min

### STANDARDS

S7L2

### Materials (per group)

- Choice of: modeling clay, a sandwich bag + small candies/beads, or a paper plate + craft scraps
- Labels or sticky notes, markers, the organelle table below

### Student Instructions

1. Decide whether you are modeling an animal or a plant cell.
2. Include and label these organelles: nucleus, cytoplasm, cell membrane, mitochondria, lysosome, vacuole. Plant cells add: cell wall, chloroplasts, large central vacuole.
3. On each label, write the organelle's job in your own words.
4. Be ready to explain why your cell "works as a system."

### TEACHER NOTES

**Quick rubric:** all required organelles present (50%), correct jobs (30%), neat & clearly labeled (20%). Encourage edible models (cookie + candy) for engagement — but no eating in lab.

### ANSWER KEY

Required animal organelles: nucleus, cytoplasm, membrane, mitochondria, lysosome, vacuole. Plant adds: cell wall, chloroplasts, large central vacuole.

## 2 Cheek vs. Onion Cells

LAB

### OBJECTIVE

Compare real animal and plant cells under a microscope.

### TIME

35–45 min

### STANDARDS

S7L2

### Materials (per group)

- Microscope, slides & cover slips
- Onion skin (thin layer), iodine or methylene blue stain
- Cotton swab for a cheek sample (teacher discretion), prepared slides as backup

### Student Instructions

1. Prepare an onion-skin slide; add a drop of stain. View at low then high power.
2. Prepare/observe a cheek-cell slide the same way.
3. Sketch each cell. Label any organelles you can see (nucleus, membrane; cell wall in onion).
4. Note: which cell has a clear, straight cell wall? Which is more rounded?

### ! TEACHER NOTES

**Safety:** stains can stain skin/clothes; goggles on. Cheek swabs are optional — use only the student's own swab and dispose properly, or substitute prepared slides. **Key observation:** onion (plant) cells are boxy with a cell wall; cheek (animal) cells are round with no wall.

### Cheek vs. Onion — Data Sheet

Name: \_\_\_\_\_ Period: \_\_\_\_\_

|                      | Onion (plant) | Cheek (animal) |
|----------------------|---------------|----------------|
| Sketch (label parts) |               |                |
| Shape                |               |                |
| Cell wall? (Y/N)     |               |                |
| Organelles seen      |               |                |

Conclusion: What is one difference you SAW between plant and animal cells?

### 3 Cell City Analogy

LAB

#### OBJECTIVE

Compare a cell to a working city to show that organelles cooperate as a system.

#### TIME

30 min

#### STANDARDS

S7L2

#### Student Instructions

A cell works like a city — each part has a job and depends on the others. Match each organelle to a city part and explain WHY it fits.

#### Cell City — Data Sheet

Name: \_\_\_\_\_ Period: \_\_\_\_\_

| Organelle         | City part I chose | Why it fits (the job) |
|-------------------|-------------------|-----------------------|
| Nucleus           |                   |                       |
| Cell membrane     |                   |                       |
| Mitochondria      |                   |                       |
| Cytoplasm         |                   |                       |
| Lysosome          |                   |                       |
| Cell wall (plant) |                   |                       |

What does it mean to say a cell is a "system"?

#### ANSWER KEY

Sample matches: nucleus = city hall/mayor; membrane = border & security gate; mitochondria = power plant; cytoplasm = roads; lysosome = recycling/waste crew; cell wall = city walls. Accept any logical match with a job-based reason.

## 4 Animal vs. Plant Cell

### WORKSHEET

#### OBJECTIVE

Identify which organelles are shared and which are plant-only.

#### TIME

15 min

#### STANDARDS

S7L2

Name: \_\_\_\_\_ Period: \_\_\_\_\_

Part A: Write BOTH, ANIMAL ONLY, or PLANT ONLY for each organelle.

| Organelle     | Found in... | Organelle             | Found in... |
|---------------|-------------|-----------------------|-------------|
| Nucleus       |             | Cell wall             |             |
| Mitochondria  |             | Chloroplast           |             |
| Cell membrane |             | Large central vacuole |             |
| Lysosome      |             | Cytoplasm             |             |

Part B: Short answer.

1. Which organelle lets a plant make its own food, and what is that process called? \_\_\_\_\_
2. Why can a plant stand up straight but an animal cell can't hold a fixed shape? \_\_\_\_\_

#### ANSWER KEY

Part A: Nucleus, mitochondria, membrane, lysosome, cytoplasm = BOTH. Cell wall, chloroplast, large central vacuole = PLANT ONLY. Part B: 1) chloroplast / photosynthesis; 2) the rigid cell wall gives plants shape & support.

## 5 Levels of Organization

### WORKSHEET

#### OBJECTIVE

Order the levels from cell to organism and give examples.

#### TIME

15 min

#### STANDARDS

S7L2

Name: \_\_\_\_\_ Period: \_\_\_\_\_

Part A: Number the levels 1 (simplest) to 5 (most complex).

Organ \_\_\_\_ Cell \_\_\_\_ Organism \_\_\_\_ Organ System \_\_\_\_ Tissue \_\_\_\_

Part B: Give an example of each level (heart pathway).

| Level        | Example |
|--------------|---------|
| Cell         |         |
| Tissue       |         |
| Organ        |         |
| Organ System |         |
| Organism     |         |

Part C:

Why can't a single cell run an entire body on its own?

#### ANSWER KEY

Part A: Organ 3, Cell 1, Organism 5, Organ System 4, Tissue 2. Part B example: muscle cell → muscle tissue → heart → cardiovascular system → human. Part C: a cell does only small jobs; complex life needs many specialized cells working together at higher levels.

## 6 Body Systems Match-Up

### WORKSHEET

#### OBJECTIVE

Match each body system to its job and key organs, and explain an interaction.

#### TIME

20 min

#### STANDARDS

S7L2

Name: \_\_\_\_\_ Period: \_\_\_\_\_

Part A: Fill in the job and a key organ.

| System         | Main Job | Key Organ |
|----------------|----------|-----------|
| Cardiovascular |          |           |
| Respiratory    |          |           |
| Digestive      |          |           |
| Excretory      |          |           |
| Nervous        |          |           |

Part B: Systems working together.

1. When you run, name TWO systems that must cooperate and what each does. \_\_\_\_\_
2. Which two systems team up to carry oxygen to your cells? \_\_\_\_\_

#### ANSWER KEY

Part A: Cardiovascular—pumps blood—heart; Respiratory—gas exchange—lungs; Digestive—breaks down food—stomach; Excretory—removes liquid waste—kidneys; Nervous—signals/control—brain. Part B: 1) e.g., muscular moves legs + respiratory pulls in O<sub>2</sub>; 2) respiratory + cardiovascular.

## 7 Writing Prompts

### WRITING

#### OBJECTIVE

Use scientific vocabulary to explain cells and systems in writing.

#### TIME

15–30 min each

#### STANDARDS

Cross-cutting

#### Level 1 — Beginner (1 paragraph)

1. **Organelle Day Off.** Pick one organelle and describe what would go wrong in the cell if it stopped working for a day.
2. **Plant or Animal?** You look at a mystery cell and see a cell wall and chloroplasts. Explain how you know what kind of cell it is.

#### Level 2 — Intermediate (3 paragraphs)

3. **Cell as a Factory.** Extend the factory comparison from the textbook: describe how at least four organelles work together to keep the "factory" running.
4. **Levels Up.** Trace one muscle cell all the way up to a whole organism, explaining what changes at each level.

#### Level 3 — Advanced (essay)

5. **One Action, Many Systems.** Choose an activity (running, eating, recovering from a cold). Argue that no single body system could do it alone, using at least three systems and how they interact.



## 8 Classroom Discussion Questions

### DISCUSSION

#### OBJECTIVE

Practice scientific reasoning through think-pair-share.

#### TIME

Warm-ups, exit tickets, seminar

#### STANDARDS

Cross-cutting

#### TEACHER NOTES

Use "think-pair-share": 30 sec think, 1 min discuss, then call on groups. Push students to defend answers with chapter evidence.

#### Quick Warm-Ups (1–2 min)

1. Is a cell alive? Which characteristics of life does it show?
2. Why is the cell membrane like a security guard?
3. Could an animal cell survive without mitochondria? Why?
4. Why don't animal cells have chloroplasts?
5. What level comes between a cell and an organ?

#### Deeper Discussions (5–15 min)

6. The textbook says a cell is a "system." What does that word mean, and what would happen if one part failed?
7. If you damaged one organ system, which others would be affected? Trace a chain of effects.
8. Plants and animals share most organelles. Why do you think the few differences (cell wall, chloroplast) matter so much?
9. Your nervous system can move signals at ~268 mph. Why is that speed useful for survival?
10. Are humans just a "bag of cells," or something more? Defend your answer using levels of organization.

## 9 Body System Public-Health Poster

GROUP

### OBJECTIVE

Research one body system and teach the class how to keep it healthy.

### TIME

2–3 class periods

### STANDARDS

S7L2

### Project Description

Each group is assigned one body system. They create a poster (or slide) that explains the system AND gives advice on keeping it healthy, then present to the class to build a "whole-body health guide."

### Poster Requirements

#### The Science

- System name & main job
- Key organs (with a labeled diagram)
- One other system it works with, and how

#### The Health Angle

- 2 habits that keep it healthy
- 1 thing that can harm it
- One "did you know?" fact

### Presentation Day

Each group presents for 2–3 minutes. Classmates fill a "Body Systems Quest" sheet noting one health tip per system.

### TEACHER NOTES

**Systems to assign:** cardiovascular, respiratory, digestive, excretory, muscular, nervous, immune. **Grading:** accuracy 40%, health advice 20%, design 20%, presentation 20%.

## 10 Vocabulary Bingo

REVIEW

### OBJECTIVE

Review chapter vocabulary in a low-stress game.

### TIME

20–30 min

### STANDARDS

S7L2

### Setup

Blank 5×5 grid per student; word list on the board; FREE middle. Read a clue, students mark the word. First to 5 in a row wins.

### Vocabulary List (24 words)

|                |              |                    |              |               |
|----------------|--------------|--------------------|--------------|---------------|
| cell           | organelle    | nucleus            | cytoplasm    | cell membrane |
| cell wall      | mitochondria | chloroplast        | lysosome     | vacuole       |
| photosynthesis | tissue       | organ              | organ system | organism      |
| cardiovascular | respiratory  | digestive          | excretory    | muscular      |
| nervous        | immune       | system interaction | homeostasis  | —             |

### Sample Clue Cards (read aloud, don't show)

| Clue to read aloud                                    | Answer             |
|-------------------------------------------------------|--------------------|
| The control center of the cell that holds DNA         | nucleus            |
| The powerhouse that releases energy from food         | mitochondria       |
| The stiff outer layer found only in plant cells       | cell wall          |
| Green organelle that captures sunlight                | chloroplast        |
| Outer border that controls what enters & leaves       | cell membrane      |
| A group of similar cells doing the same job           | tissue             |
| Different tissues working together, like the heart    | organ              |
| The system that pumps blood                           | cardiovascular     |
| The system that brings in oxygen                      | respiratory        |
| The cleanup organelle that breaks down waste          | lysosome           |
| Making food from sunlight, water, and CO <sub>2</sub> | photosynthesis     |
| Two systems working together for a life process       | system interaction |

### TEACHER NOTES

**Easy adaptation:** give struggling readers the list WITH definitions during play. Run multiple patterns (line, X, blackout).